CYBERSECURITY & ETHICAL HACKING INTERNSHIP

Task 2 — Lab Build Report

Building a Personal, Isolated Cybersecurity Practice Lab

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LAB COMPONENTS	Kali Linux (Attacker VM) + OWASP Juice Shop (Target)
VIRTUALIZATION	Oracle VirtualBox
NETWORK DESIGN	NAT (Internet/Updates) + Host-Only (Isolated Lab Traffic)
VALIDATION TOOLS	Nmap, Wireshark, Burp Suite

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1. Objective

The objective of this task was to design and build a safe, fully isolated cybersecurity practice environment on a personal computer. This hands-on lab forms the foundation for all subsequent tasks in the internship track, including vulnerability assessment, web application testing, exploitation practice, and incident response.

The lab was built with the following components:

- One Attacker machine — Kali Linux
- One vulnerable target application — OWASP Juice Shop, deployed via Docker
- Proper network segmentation using NAT (for internet access/updates) and a Host-Only network (for isolated lab traffic)

1.1 What This Task Delivers

Skill Area	Practical Outcome
Virtualization	Running multiple isolated systems for pentesting
Lab isolation	Preventing real-world system damage
Web targets setup	Deploying an intentionally vulnerable application
Basic networking	NAT vs. Host-Only and subnet usage
Tool awareness	Nmap, Burp Suite and Wireshark validation

1.2 Scope and Boundaries

To keep the build focused and reproducible, the following scope was applied:

- In scope: local virtualization, dual-network configuration, target deployment, and validation using Nmap, Wireshark, and Burp Suite.
- Out of scope: any scanning, exploitation, or traffic capture directed at systems outside the lab's own Host-Only network.
- The target application, OWASP Juice Shop, was chosen because it is intentionally vulnerable and explicitly designed for this kind of legal, controlled practice.

2. Lab Architecture

The lab uses a two-network design so that the attacker and target machines can communicate with each other on a private, host-only segment, while still retaining a separate NAT-based path to the internet for patching and package installation. This mirrors the segmentation used in real penetration-testing labs, where test traffic never touches the production or public network.

```
|     - NAT        -> Internet / Updates  |
|     - Host-Only  -> Isolated Lab Traffic|
+-----------------------------------+
```

On the deployed lab, the Kali Linux attacker VM was configured with two network adapters — Adapter 1 attached to NAT for outbound internet access, and Adapter 2 attached to a Host-Only network (vboxnet0) for direct, isolated communication with the target application. Both were confirmed active and connected in the VirtualBox settings shown in Section 5.2.

This separation is deliberate. If the attacker VM used only NAT, its lab traffic would be routed and potentially exposed alongside normal internet traffic; if it used only Host-Only networking, it would have no route out to fetch updates or new tools. Running both adapters together gives the attacker machine a private, addressable lab segment for testing while still behaving like a normal, patchable Linux system.

2.1 NAT vs. Host-Only — Design Comparison

Aspect	NAT Adapter	Host-Only Adapter
Internet access	Yes — used for updates/downloads	No — fully isolated
Visible to the host network	No — traffic is translated by VirtualBox	No — private virtual switch only
Attacker ↔ Target reachability	Not used for lab traffic	Yes — primary channel for testing
Risk if misused	Low, standard outbound traffic	None — cannot reach production systems
Used in this lab for	apt updates, tool downloads	Nmap, Wireshark, Burp Suite traffic

3. Prerequisites

Before beginning the build, the following host system requirements and prerequisites were reviewed to ensure the lab would run smoothly:

3.1 Host System Requirements

Requirement	Recommendation
RAM	Minimum 8 GB, preferably 16 GB
Disk Space	40–60 GB free
CPU	Virtualization (VT-x / AMD-V) enabled
Operating System	Windows / macOS / Linux

3.2 Software and Tools Used

Tool	Purpose
Oracle VirtualBox	Virtualization platform hosting both VMs

Tool	Purpose
Kali Linux VM	Attacker machine with pre-installed security tools
OWASP Juice Shop	Intentionally vulnerable web application (target)
Docker	Containerized deployment of Juice Shop
Nmap	Host discovery and connectivity validation
Wireshark	Network traffic capture and packet analysis
Burp Suite	Web proxy for intercepting and inspecting HTTP traffic

3.3 Risk and Isolation Considerations

Because the target application is intentionally vulnerable, a few precautions were treated as mandatory rather than optional before any testing traffic was sent:

- The target container is only ever reachable from the Host-Only network — it is never exposed on the NAT adapter or on the physical host network.
- No port-forwarding rules were created that would expose the Juice Shop container to the internet.
- The attacker VM's snapshots are taken before any exploitation activity, so the lab can be reset to a clean state if needed.
- All scanning and interception in this report targeted only machines inside the lab's own 192.168.56.0/24 subnet.

4. Build Methodology

The lab was assembled in five sequential stages, following the step-by-step instructions provided in the task brief. Each stage was validated before moving on to the next, so that any misconfiguration could be isolated and corrected early rather than compounding later in the build.

Stage	Action	Outcome
1	Install VirtualBox; create a Host-Only network; enable NAT	Virtualization platform ready with both required network types available
2	Deploy the Kali Linux attacker VM (2 CPUs, 3–4 GB RAM, dual NIC)	Attacker VM boots with NAT + Host-Only adapters attached
3	Deploy the target (OWASP Juice Shop via Docker)	Vulnerable web application running and exposed on port 3000
4	Verify connectivity (`nmap -sn` across the lab subnet)	Attacker machine confirmed to see the target on the isolated network
5	Basic validation (browser access, Burp interception, Nmap scan)	End-to-end tooling confirmed functional against the target

Working folders were also created on the attacker VM to keep scan output, notes, packet captures and screenshots organised for this and future tasks:

```
sudo apt update && sudo apt upgrade -y
mkdir -p ~/lab/{scans,notes,pcaps,screenshots}
```

5. Environment Setup

5.1 Attacker Machine — Kali Linux VM

The Kali Linux attacker virtual machine was imported and deployed inside Oracle VirtualBox as an ARM64 guest, allocated its own virtual disk, VirtioSCSI storage controller, and dedicated display and audio devices. The completed VM configuration (left) confirms the guest OS, storage, and dual network adapters. Once powered on, the desktop environment (right) loaded successfully, confirming the VM boots correctly and is ready for tool usage.

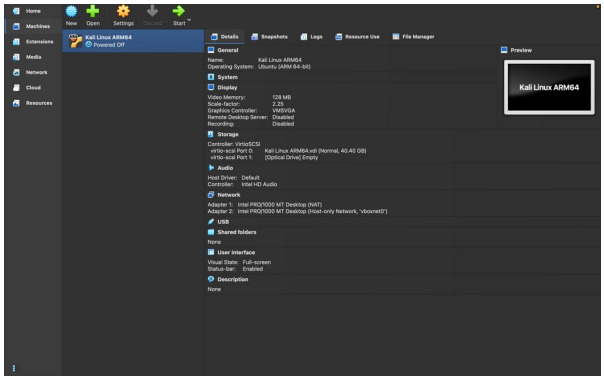

Figure 1 — VirtualBox machine summary for the Kali Linux ARM64 attacker VM.

Figure 2 — Kali Linux desktop running inside VirtualBox.

5.2 Network Adapter Configuration

Two virtual network adapters were configured on the attacker VM in line with the required NAT + Host-Only design: Adapter 1 was attached to NAT, providing outbound internet access for updates and package installation, while Adapter 2 was attached to a Host-Only network (vboxnet0), providing an isolated segment for direct communication with the target machine.

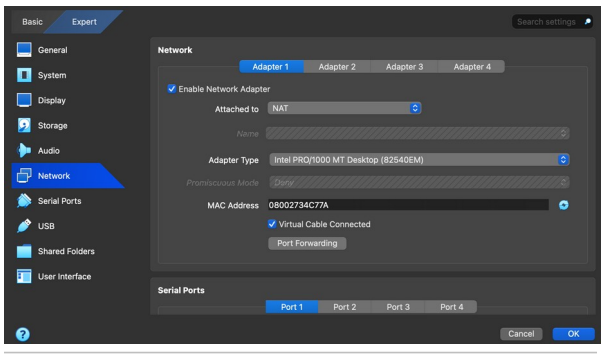

Figure 3 — Adapter 1 configured with NAT.

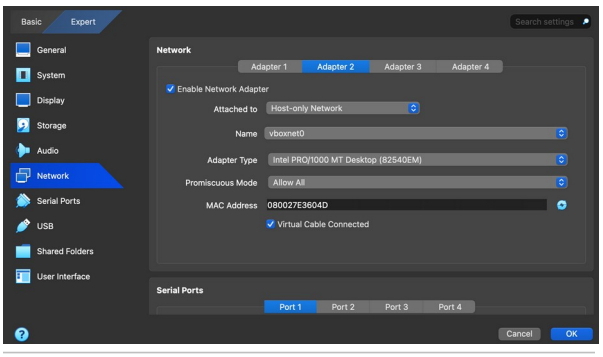

Figure 4 — Adapter 2 configured with Host-Only (vboxnet0).

5.3 Verifying Interfaces on the Kali VM

Inside the running Kali Linux VM, the `ip a` command was used to confirm both network interfaces were active and correctly addressed — the NAT interface (eth0) received a 10.0.2.15/24 address, while the Host-Only interface (eth1) was also up, alongside the Docker bridge interface (docker0) used to expose the target container.

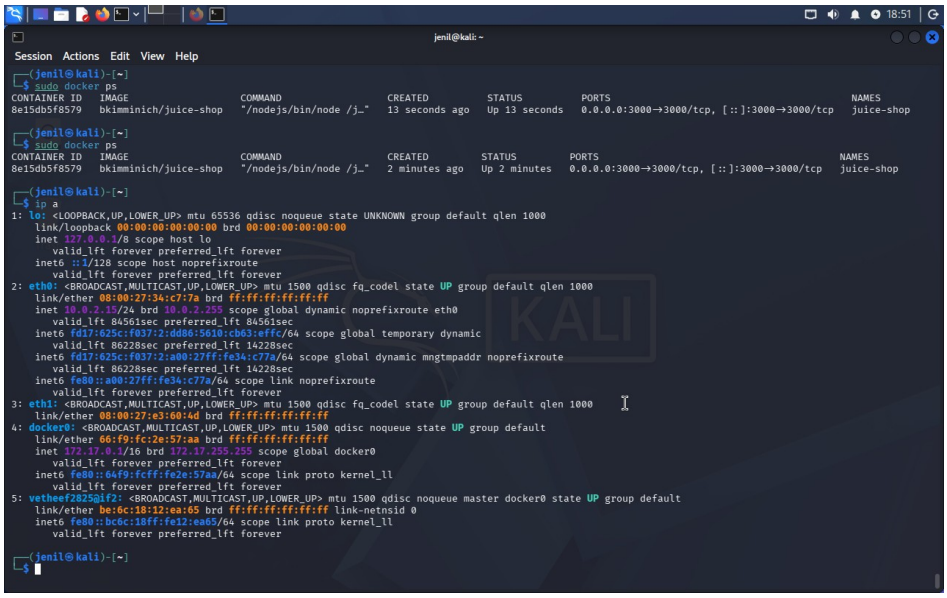

Figure 5 — Output of `docker ps` and `ip a`, confirming the Juice Shop container is running and both network interfaces are active.

6. Target Deployment — OWASP Juice Shop

OWASP Juice Shop was selected as the vulnerable target application. It was deployed as a Docker container directly inside the Kali Linux VM using the official image, exposing the application on port 3000:

```
sudo apt install -y docker.io
sudo systemctl enable --now docker
sudo docker pull bkimminich/juice-shop
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

The `docker ps` output (Figure 5) confirmed the container was up and running, with port 3000 mapped and available on both IPv4 and IPv6. Accessing the application from a browser on port 3000 confirmed the target was successfully deployed and reachable, with the storefront loading correctly, including its product catalogue and tracking-cookie banner.

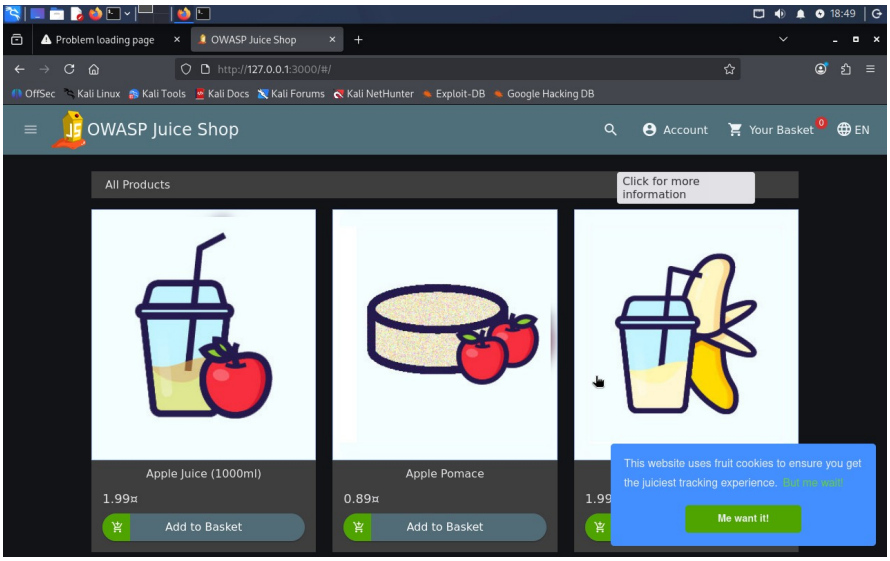

Figure 6 — OWASP Juice Shop storefront running successfully at http://127.0.0.1:3000/.

7. Connectivity Validation — Nmap

To confirm that the attacker machine could reach the target across the Host-Only network, a host-discovery sweep was run against the lab subnet:

```
nmap -sn 192.168.56.0/24
```

The scan reported all 256 addresses in the 192.168.56.0/24 range as up, with sub-millisecond latency — confirming the Host-Only network was correctly configured and that the attacker VM could see every host on the isolated lab segment, including the target machine.

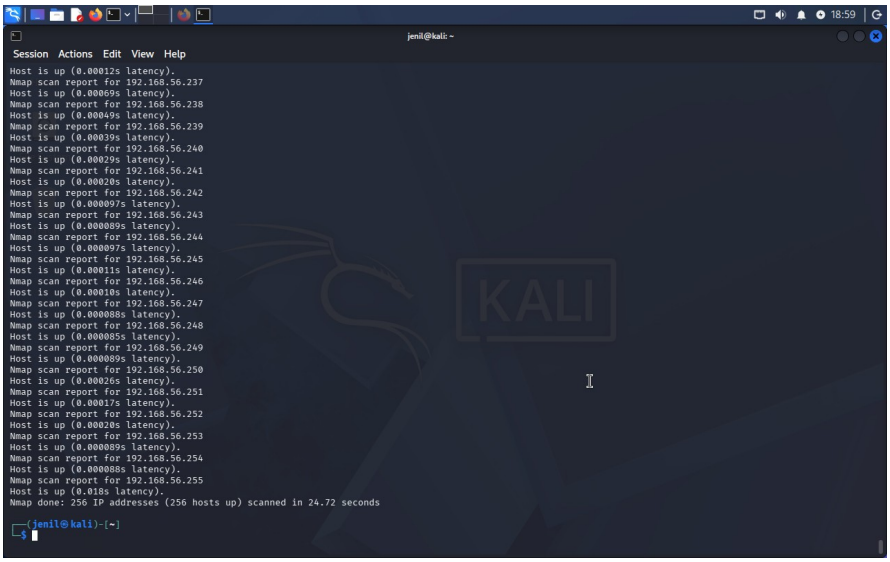

Figure 7 — Nmap host-discovery scan confirming 256 hosts reachable on the 192.168.56.0/24 Host-Only network.

8. Traffic Analysis — Wireshark

Wireshark was launched on the eth1 (Host-Only) interface to validate that lab network traffic could be captured for later analysis. The tool was confirmed to start and stop capture sessions correctly, writing capture files to /tmp for further review.

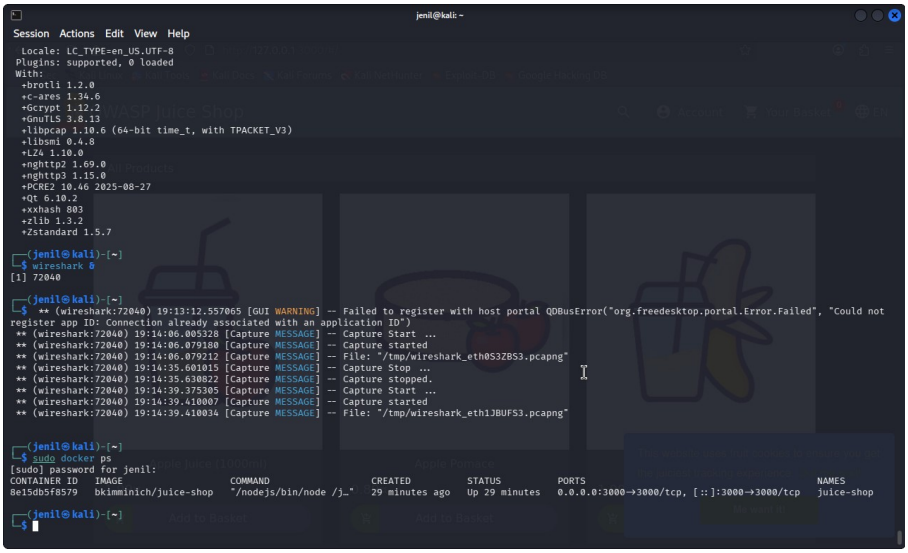

Figure 8 — Wireshark launched from the terminal, showing successful capture start/stop events and the Juice Shop container status.

A short initial capture on eth1 (left) picked up multicast DNS (mDNS) query and response traffic. The capture was then left running for a longer interval, gathering 61 packets (right) including additional MDNS and ICMPv6 Router Advertisement traffic — confirming stable, continuous packet capture on the lab network.

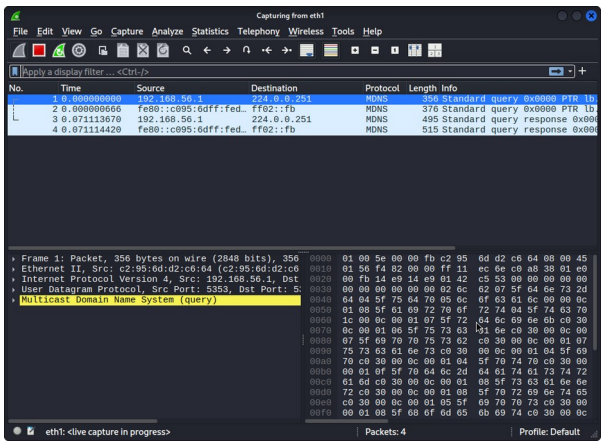

Figure 9 — Initial capture on eth1 (4 packets, MDNS).

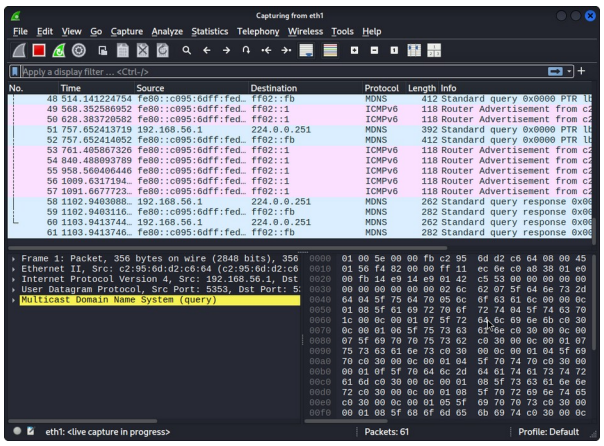

Figure 10 — Extended capture on eth1 (61 packets).

9. Web Traffic Interception — Burp Suite

Burp Suite Community Edition was configured to validate the web-proxy interception workflow that will be used in later web-application testing tasks. A proxy listener was configured on 127.0.0.1:8080, with interception rules enabled to stall requests for review, excluding common static file types (images, CSS, fonts, etc.). With the listener

active, browser traffic routed through the proxy was captured in the HTTP History tab, confirming successful interception and logging of outbound requests.

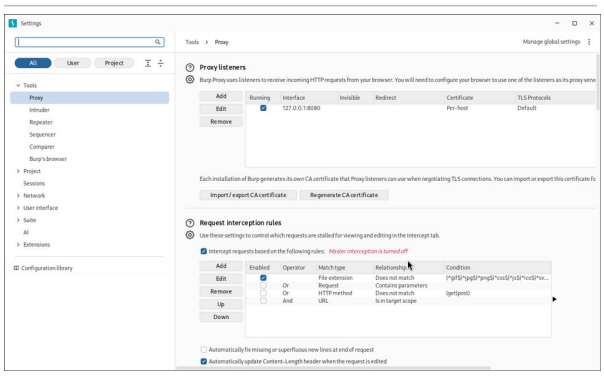

Figure 11 — Burp Suite Proxy listener on 127.0.0.1:8080 and interception rules.

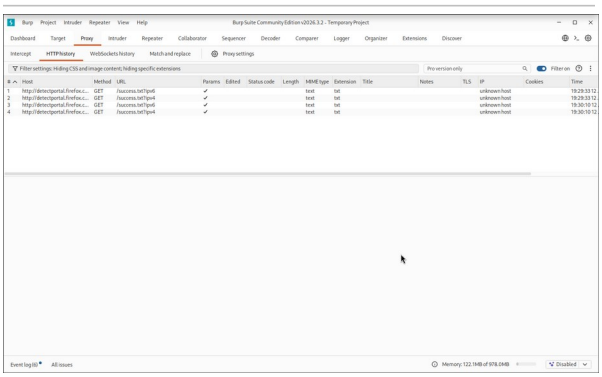

Figure 12 — Burp Suite HTTP History showing intercepted requests.

10. Evidence Summary

10.1 Virtual Machine Inventory

VM / Component	Role	OS / Image	Status
Kali Linux ARM64	Attacker	Kali Linux (Ubuntu ARM 64-bit base)	Running
juice-shop	Target (container)	bkimminich/juice-shop	Up, port 3000 mapped

10.2 IP / Network Configuration

Interface	Network	Address	Purpose
eth0	NAT	10.0.2.15/24	Internet access & package updates
eth1	Host-Only (vboxnet0)	192.168.56.x/24	Isolated attacker-to-target lab traffic
docker0	Docker bridge	172.17.0.1/16	Container networking for Juice Shop

10.3 Validation Checklist

Validation Step	Tool	Result
Host-discovery across lab subnet	Nmap (-sn)	256/256 hosts reachable
Target application reachable in browser	Firefox	Juice Shop storefront loaded

Validation Step	Tool	Result
Packet capture on Host-Only interface	Wireshark	61 packets captured (MDNS / ICMPv6)
Proxy interception of browser traffic	Burp Suite	Requests logged in HTTP History

10.4 Screenshot Index

Figure	Description
1 – 2	Kali Linux ARM64 VM summary and running desktop
3 – 4	NAT (Adapter 1) and Host-Only (Adapter 2) network configuration
5	`docker ps` and `ip a` output confirming interfaces and container status
6	OWASP Juice Shop storefront reachable in the browser
7	Nmap host-discovery sweep across the lab subnet
8	Wireshark launched and capturing on eth1
9 – 10	Initial (4-packet) and extended (61-packet) Wireshark captures
11 – 12	Burp Suite proxy listener configuration and HTTP History

11. Challenges and Observations

- Ensuring the Host-Only adapter (vboxnet0) was enabled and had a virtual cable connected before boot was necessary for the attacker VM to see the target network.
- Running Docker commands required elevated privileges (sudo), so the Juice Shop container was deployed and inspected using an authenticated shell session.
- Wireshark needed to be pointed explicitly at the eth1 interface to capture Host-Only lab traffic rather than the NAT-facing eth0 interface.
- Burp Suite's default interception rule excludes common static file extensions, which kept the HTTP History focused on meaningful application requests.
- The initial Nmap sweep took under 25 seconds to enumerate a full /24, which was fast enough to re-run after any network change to re-confirm reachability.
- Because both VM adapters share the same physical host, the ARM64 Kali image needed the VirtioSCSI storage controller for stable disk performance during setup.

12. Recommendations and Next Steps

With the base lab validated, the following actions are recommended before moving into the next task in the internship track:

- Take a VirtualBox snapshot of both the attacker VM and the target container's known-good state, so the environment can be reset quickly after a failed exploitation attempt.
- Import the Burp Suite CA certificate into the browser used for testing, to allow HTTPS interception in addition to the HTTP traffic captured in Section 9.

- Extend the Nmap validation from a host-discovery sweep (`-sn`) to a service/version scan (`-sV -sC`) against the target once vulnerability assessment begins.
- Save Wireshark captures directly into the `~/lab/pcaps` working folder created in Section 4, using descriptive filenames tied to each testing session.
- Keep the Host-Only network as the only path used for actual attack traffic against the target, reserving the NAT adapter strictly for updates.

13. References and Learning Resources

The following official documentation and beginner-friendly resources were used as references while building and validating this lab:

Resource	Purpose
VirtualBox — virtualbox.org/wiki/Downloads	Official download and setup for the virtualization platform
Kali Linux VM — kali.org/get-kali	Official prebuilt Kali Linux virtual machine images
OWASP Juice Shop — github.com/juice-shop/juice-shop	Source and documentation for the target application
Docker Engine — docs.docker.com/engine/install	Installation guide for the container runtime used to deploy Juice Shop
Burp Suite Documentation — portswigger.net/burp/documentation/desktop	Reference for proxy configuration and interception rules
Wireshark Documentation — wireshark.org/docs	Reference for packet capture and protocol analysis

14. Conclusion

This task successfully established a safe, segmented, and reproducible personal cybersecurity lab. A Kali Linux attacker VM and an OWASP Juice Shop target container were deployed inside Oracle VirtualBox, connected through a dual NAT + Host-Only network design that keeps lab traffic fully isolated from the host's production network while still allowing internet access for updates.

Connectivity and tooling were validated end-to-end: Nmap confirmed host discovery across the lab subnet, the Juice Shop application was reachable and functional in the browser, Wireshark successfully captured live traffic on the isolated interface, and Burp Suite was configured and verified as an intercepting proxy. With this foundation in place, the environment is ready to support the subsequent tasks in the internship track, including vulnerability assessment, web application testing, exploitation practice, and incident response.

14.1 Key Takeaways

- A dual-adapter (NAT + Host-Only) design is sufficient to keep a vulnerable target fully isolated while still allowing the attacker VM to stay updated.
- Docker significantly simplifies target deployment — the Juice Shop application was up and reachable within minutes of installing the Docker engine.
- Validating each layer independently — network reachability (Nmap), packet visibility (Wireshark), and proxy interception (Burp Suite) — makes it much easier to isolate problems before more advanced testing begins.

15. Appendix — Deliverables

The following deliverables were produced as part of this task:

- Full Lab Build report (this document), including annotated screenshots
- Evidence folder containing:
  - VM list
  - IP configuration
  - Ping / Nmap validation output
  - Burp Suite and Wireshark screenshots

— End of Report —